9-6-24

When solving ODE's:

First-order
- Separation of variables
- when linear: integrating factor

2nd-order
linear, constant coeff
Assume sol'n of form $e^{\lambda t}$ and set charact. eq'n ...

Or
Guess + check

e.g.
$y' = -2y$
$\boxed{y = c_1 e^{-2x}}$, $c_2 e^{2x}$
check: $-2c_1 e^{-2x}$ $-2y = -2c_1 e^{-2x}$

$y'' + y = 0$
$y = A\sin(x) + B\cos(x)$

Must check.

ex) Farlow p. 207

$$z_x + z_y + 2z = 0 \qquad IC \quad z(x,0) = \sin(x)$$

① Param. The IC if needed $\qquad \langle \tau, 0, \sin(\tau) \rangle$

$\langle x_0(\tau), y_0(\tau), z_0(\tau) \rangle$

② Identify P, Q, R

$P = 1 \qquad Q = 1 \qquad R = -2z$

③ check conditions for existence + uniqueness

need to show for HW#4 → (i) $P, Q, R \in C^1$? Yes $\quad x_0(\tau), y_0(\tau), z_0(\tau) \in C^1$? Yes

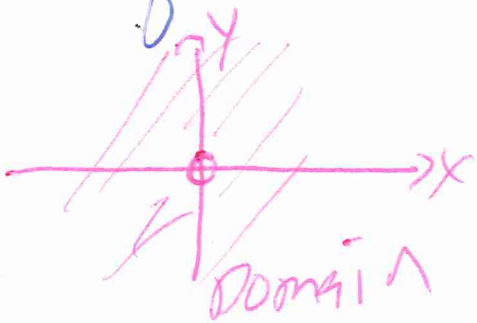
(ii) I.C. charact?

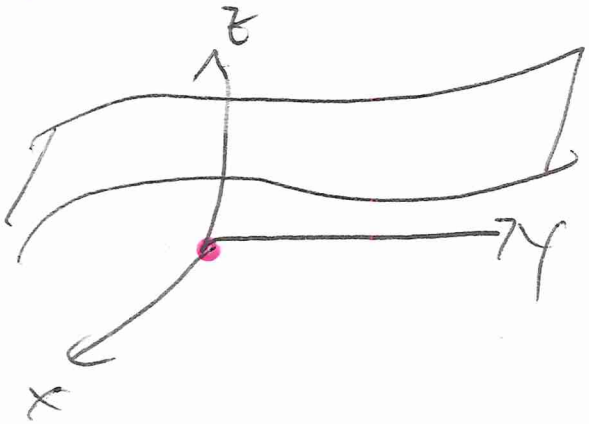
NO

$$\begin{vmatrix} P|_{IC} & Q|_{IC} \\ x_0' & y_0' \end{vmatrix} = \begin{vmatrix} 1 & 1 \\ 1 & 0 \end{vmatrix} = -1 \neq 0$$

$= \tau$

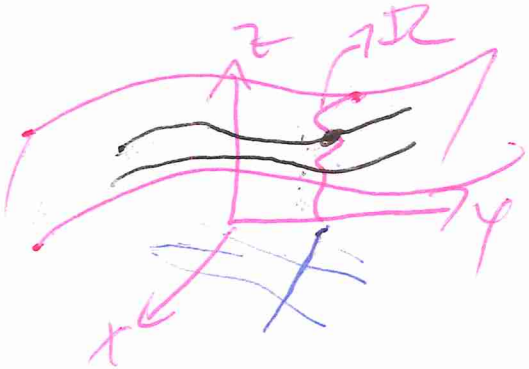
$\Rightarrow$ we have a unique sol'n :)





④ Find the characteristic curves

$$\frac{dx}{ds} = P \qquad \frac{dy}{ds} = Q \qquad \frac{dz}{ds} = R$$



$\cancel{x_0(s) = x_0}$

$x_0(s=0) = x_0(\tau) \qquad y_0(s=0) = y_0(\tau) \qquad z_0(s=0) = z_0(\tau)$

solve treating τ as a constant.

$$\frac{dx}{ds} = 1 \qquad \frac{dy}{ds} = 1 \qquad \frac{dz}{ds} = -2z$$

$x_0(s=0) = \tau \qquad y(s=0) = 0 \qquad z(s=0) = \sin(\tau)$

$dx = ds \qquad dy = ds \qquad \frac{dz}{z} = -2\,ds$

$x = s + c_1 \qquad y = s + c_2 \qquad \ln|z| = -2s + c_3$

$s=0 \quad x = \tau \qquad s=0 \quad y=0 \qquad z = c_3 e^{-2s}$

$\tau = 0 + c_1 \Rightarrow c_1 = \tau \qquad \Rightarrow c_2 = 0 \qquad s=0 \quad z = \sin(\tau)$

$\sin(\tau) = c_3 e^0 = c_3$

$\underline{x = s + \tau}$ ① $\qquad \underline{y = s}$ ② $\qquad \underline{z = \sin(\tau) e^{-2s}}$ ③

⑤ eliminate s, τ so that sol'n $z = f(x,y)$

$s = y$ ②

① $x = y + \tau \Rightarrow \tau = x - y$

③ $z = \sin(\tau)\, e^{-2s} = \boxed{\sin(x-y)\, e^{-2y} = z}$

⑥ check $z_x + z_y + 2z = 0$

$z_x = \cos(x-y)\, e^{-2y}$

$z_y = -\cos(x-y)\, e^{-2y}$ $\boxed{-2\sin(x-y)\, e^{-2y}}$

$2z = \boxed{2\sin(x-y)\, e^{-2y}}$

$= 0$ ☺

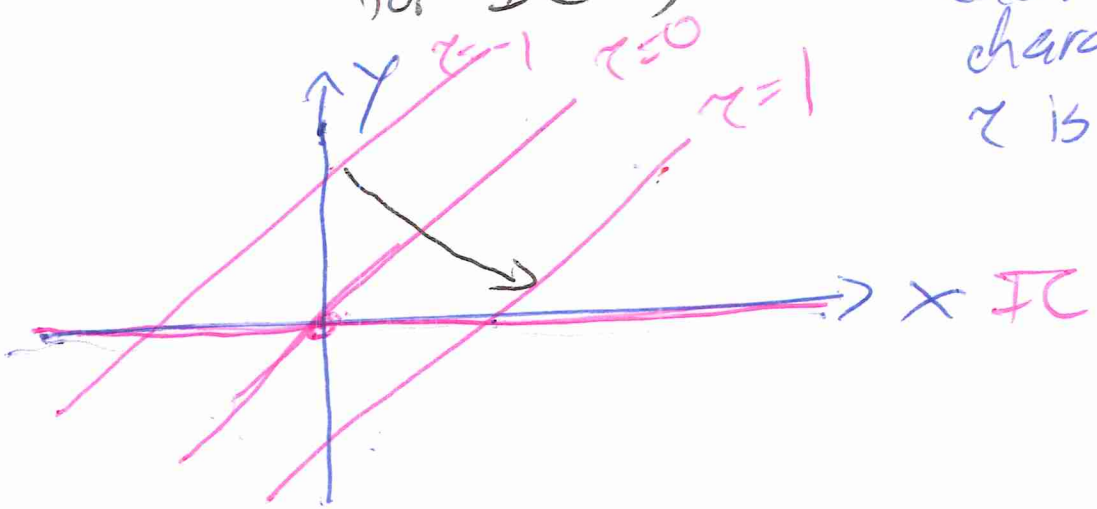
⑦ To help visualize soln, is to graph <u>characteristics</u> $x(s,\tau), y(s,\tau)$

$x = s+\tau \qquad y = s$

eliminate s (because τ is the param for IC)

$x = y+\tau$

$\boxed{y = x-\tau}$



each charact, τ is fixed

Recall IC $\quad z(\underbrace{x,0}_{y=0!}) = \sin(x) \qquad x=\tau$

soln is $z = \sin(\tau)e^{-2s}$
$= \sin(\tau)\, e^{-2y}$

Along characteristics z decays exponentially for $y \to \infty$

ex) $x z_x + (y^2+1) z_y = z \qquad z(x,0) = e^x$

① Param IC $\quad x = \tau, \; y = 0, \; z = e^\tau$

② Identify P, Q, R

$P = x \qquad Q = y^2+1 \qquad R = z$

③ check conditions for existence & uniqueness of sol'n

$P, Q, R \in C^1$? Yes, all polynomials

$x_0(\tau), y_0(\tau), z_0(\tau) \in C^1$? ~~Yes~~ Yes, all polynomials or exp.

IC charact? $\begin{vmatrix} P|_{IC} & Q|_{IC} \\ x_0'(\tau) & y_0'(\tau) \end{vmatrix} = \begin{vmatrix} x|_{IC} & y^2+1|_{IC} \\ 1 & 0 \end{vmatrix} = \begin{vmatrix} \tau & 1 \\ 1 & 0 \end{vmatrix} = -1 \neq 0$:)

(9) Find charact. curves

$$\frac{dx}{ds} = P \qquad \frac{dy}{ds} = Q \qquad \frac{dz}{ds} = R$$

$$x(s=0) = x_0(\tau) \qquad y(s=0) = y_0(\tau) \qquad z(s=0) = z_0(\tau)$$

$$\frac{dx}{ds} = x$$

$$x(s=0) = \tau$$

Guess & check

$$x = c_1 e^s$$

$$s=0, \quad x = \tau$$

$$\tau = c_1(e^0)$$

$$\Rightarrow c_1 = \tau$$

$$\underline{x = \tau e^s}$$

(1)

$$\frac{dy}{ds} = y^2+1$$

$$y(s=0) = 0$$

S of V

$$\frac{dy}{y^2+1} = ds$$

$$\operatorname{atan}(y) = s + c_2$$

$$y = \tan(s + c_2)$$

$$s=0 \quad y=0$$

$$0 = \tan(c_2) \Rightarrow c_2 = 0$$

$$\underline{y = \tan(s)}$$

(2)

$$\frac{dz}{ds} = z$$

$$z(s=0) = e^\tau$$

Guess & check

$$z = c_3 e^s$$

$$s=0, \quad z = e^\tau$$

$$e^\tau = c_3 e^0$$

$$\Rightarrow c_3 = e^\tau$$

$$z = e^\tau e^s = e^{\tau+s}$$

$$\underline{z = e^{\tau+s}}$$

(3)

⑤ Find $z = z(x,y)$

eliminate τ & s

From ② $s = \operatorname{atan}(y)$

① $x = \tau e^{\operatorname{atan}(y)}$

$\tau = x e^{-\operatorname{atan}(y)}$

From ③

$z = e^{[x e^{-\operatorname{atan}(y)} + \operatorname{atan}(y)]}$

⑥ characteristics: $x(s,\tau)$ $y(s,\tau)$ eliminate s

From ② $s = \operatorname{atan}(y)$

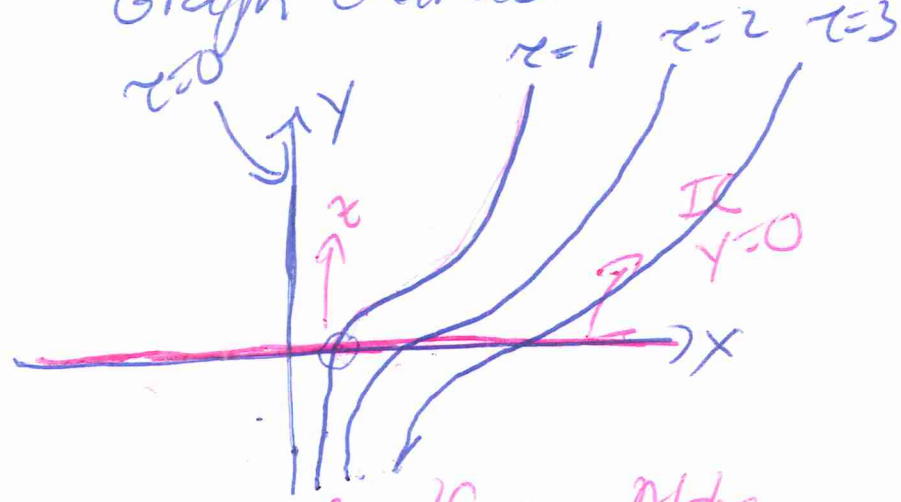
① $x = \tau e^{\operatorname{atan}(y)}$ → $\tau = 0$, $x = 0$

③ $z = e^{\tau + \operatorname{atan}(y)}$

$y = f(x)$ $e^{\operatorname{atan}(y)} = \frac{x}{\tau}$

$\operatorname{atan}(y) = \ln\left(\frac{x}{\tau}\right)$

Graph characteristics



using Wolfram Alpha

Along each charact $z = e^{\tau} e^{s}$

grows exp as $e^{\operatorname{atan}(y)}$ along each charact.